INTELLIGENT EMAIL MANAGEMENT & SPAM DETECTION PLATFORM

Database Management System — Final Review



Group No.: 13

Team Members

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PROBLEM & OBJECTIVES

PROBLEM

Managing large volumes of email requires organized storage.

Spam emails create security and usability problems.

User, email and spam information must remain connected and consistent.

Fast search and classification are required.

OBJECTIVES

Build a normalized relational database.

Manage users, emails and categories.

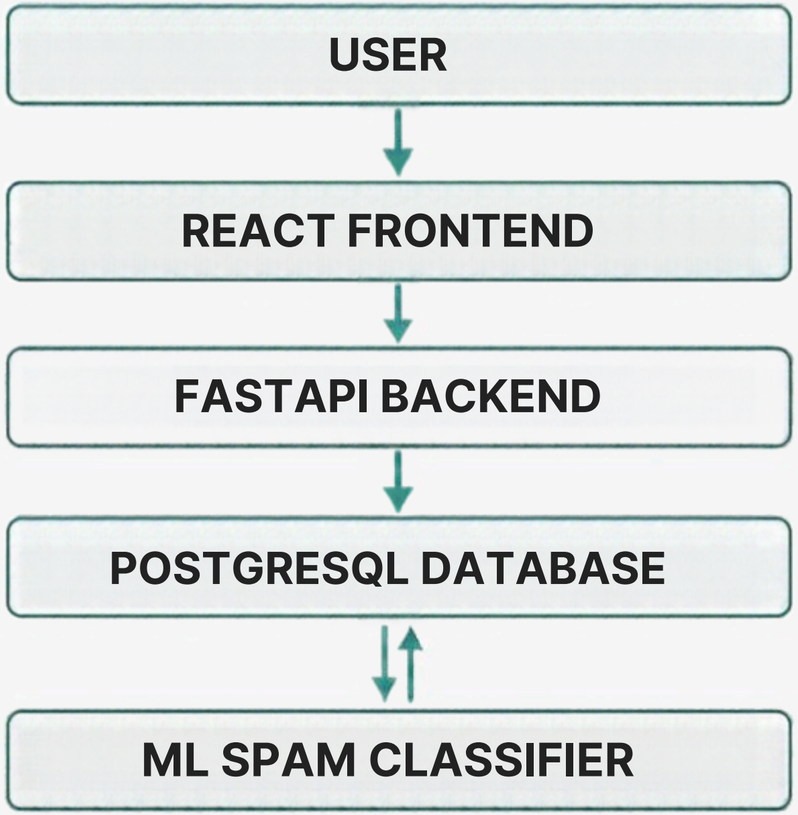
Detect and classify spam emails.

Store spam-detection results.

Provide email search and management.

Demonstrate practical DBMS features.

SYSTEM ARCHITECTURE



Frontend: React + Vite

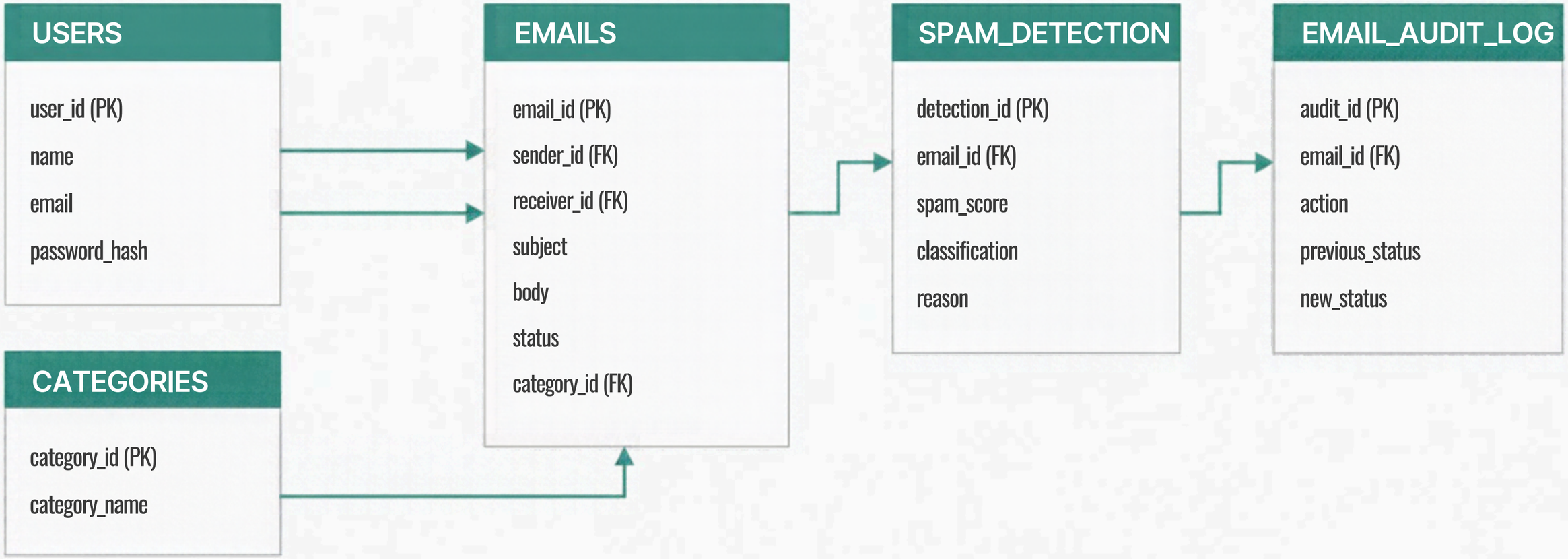
Backend: FastAPI

Database: PostgreSQL

Machine Learning: Python + scikit-learn

ORM: SQLAlchemy

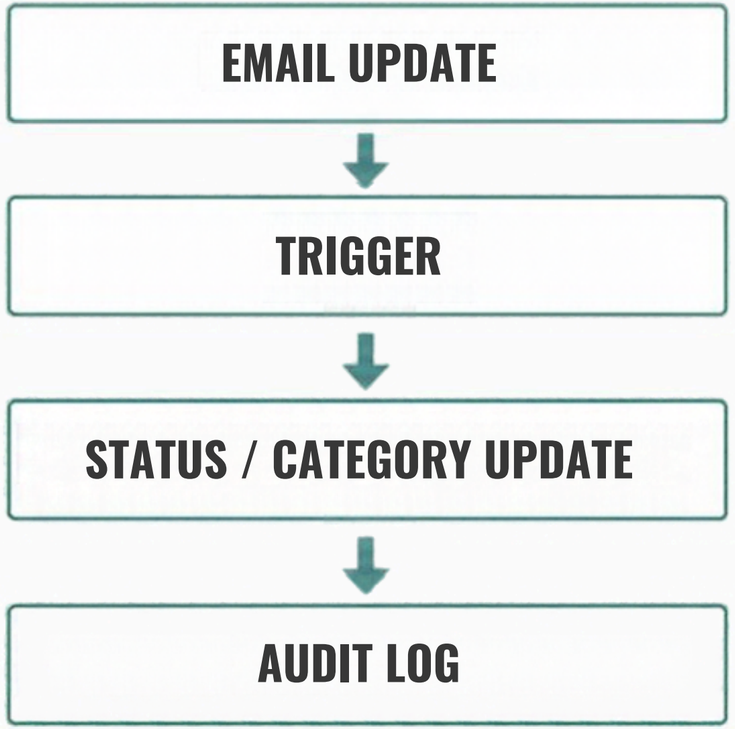
DATABASE DESIGN



Relational schema normalized to 3NF.

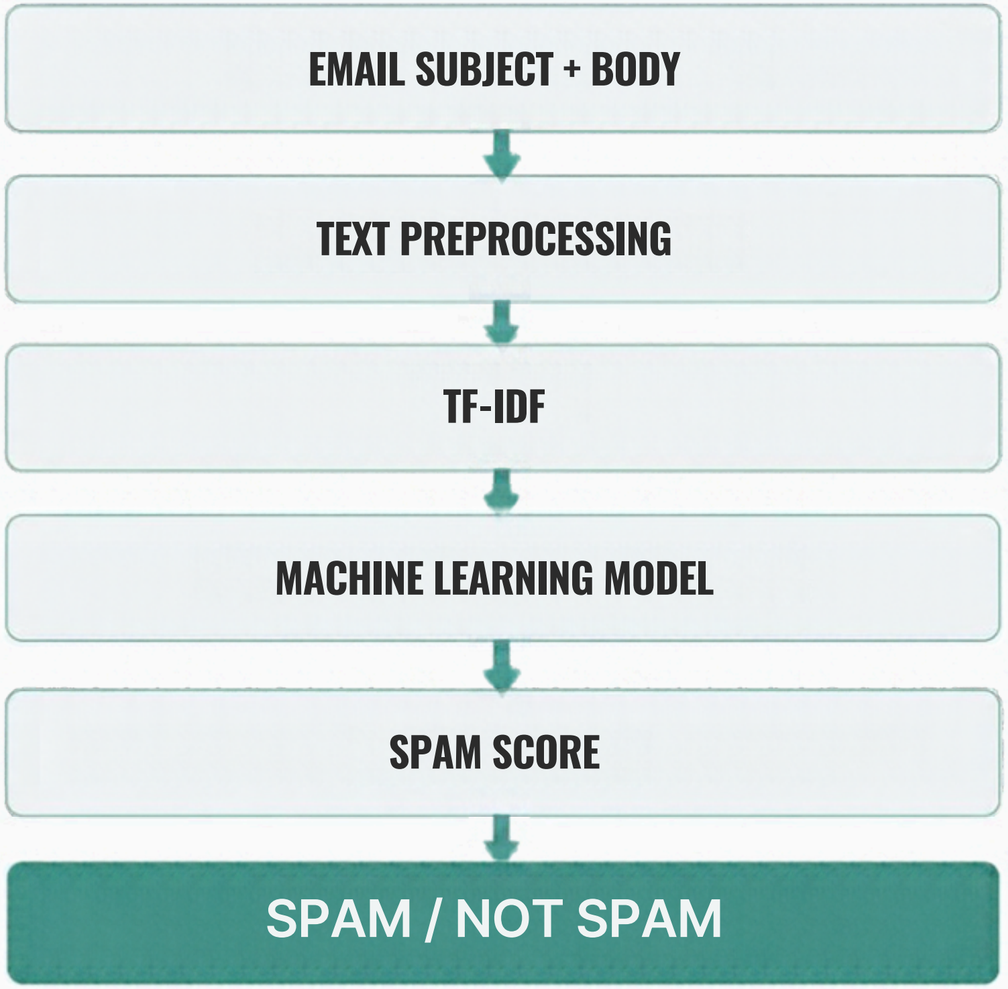
DBMS FEATURES IMPLEMENTED

- ✓ Primary and foreign keys
- ✓ NOT NULL / UNIQUE / CHECK CONSTRAINTS
- ✓ Referential integrity
- ✓
- ✓ PostgreSQL triggers
- ✓ Stored functions
- ✓ Audit logging
- ✓ SQL JOIN queries
- ✓ Aggregate queries
- ✓ Full-text search
- ✓ GIN index



	user_id [PK] inte	name character varying (100	email character varying (255)	password_hash character varying (255)	created_at timestamp with time zone
1	1	Admin Evaluator	admin@emailmgmt.local	$2b$12$NmQRzTbR1GxHFcSfr4KbJO4GzDhh64EuQi8BhYDPKgskiIqhYU...	2026-09-05 21:34:35.554378+05:...
2	2	John Doe	john.doe@emailmgmt.local	$2b$12$nuvQ.KT/ZDV/h3wO2hTokO6Zw/e0KMheL.hZByLcBpnWdU0rG...	2026-09-10 21:34:35.554378+05:...
3	3	Alice Smith	[illegible]	[illegible]	[illegible]9-15 21:34:35.554378+05:...
4	4	Bob Wilson (Guide)	bob.wilson@emailmgmt.local	$2b$12$nuvQ.KT/ZDV/h3wO2hTokO6Zw/e0KMheL.hZByLcBpnWdU0rG...	2026-09-20 21:34:35.554378+05:...
5	5	Phishing Spammer	spammer@win-cash-now.net	$2b$12$nuvQ.KT/ZDV/h3wO2hTokO6Zw/e0KMheL.hZByLcBpnWdU0rG...	2026-09-25 21:34:35.554378+05:...
6	6	Deals Marketing	deals@superdiscounts-online.xyz	$2b$12$nuvQ.KT/ZDV/h3wO2hTokO6Zw/e0KMheL.hZByLcBpnWdU0rG...	2026-09-27 21:34:35.554378+05:...

SPAM DETECTION PIPELINE



DATASET

179 synthetic emails

- 90 Legitimate
- 89 Spam

MODELS EVALUATED

Multinomial Naive Bayes

Logistic Regression

Best model selected using F1-score.

Dataset was generated for project development and testing.

SPAM CLASSIFIER RESULTS

94.12%	90.00%	100%	94.74%
Accuracy	Precision	Recall	F1 Score

Selected Model:
Logistic Regression (C = 2.0)

Test Set:
36 samples

Confusion Matrix:
TN = 16
FP = 2
FN = 0
TP = 18

	detection_id [PK] integer	email_id integer	spam_score double preci	classification character vary	reason text	detected_at timestamp with
1	1	1	0.02	Not Spam	Genuine academic correspondence containing standard institutional vocabulary.	2026-10-02 …
2	2	2	0.04	Not Spam	Legitimate technical communication regarding database schemas and project design.	2026-10-03 …
3	3	3	0.05	Not Spam	Informational departmental notification regarding seminar schedule.	2026-10-04 …
4	4	4	0.01	Not Spam	Project submission and student-advisor academic correspondence.	2026-10-04 …
5	5	5	0.06	Not Spam	Social informal peer communication regarding lunch event.	2026-10-05 …
6	6	6	0.03	Not Spam	Academic guidelines regarding thesis formatting and IEEE citation.	2026-10-05 …
7	7	7	0.98	Spam	Triggered high-risk keywords: [winner, lottery, cash prize, sweepstakes, credit card, urgent]. Exceeds spam threshold (…	2026-09-30 …
8	8	8	0.96	Spam	Identified credential phishing indicators: [urgent, bank account, suspended, verify password, permanent termination].	2026-10-01 …
9	9	9	0.92	Spam	Commercial spam patterns detected: [cheap meds, miracle pills, discount, free worldwide delivery].	2026-10-02 …
10	10	10	0.97	Spam	Financial advance-fee scam markers identified: [unclaimed inheritance, escrow funds, wire transfer, banking coordinat…	2026-10-03 …
11	11	11	0.94	Spam	High-confidence fraudulent investment scheme: [crypto investment, guaranteed return, double, risk-free profits].	2026-10-04 …
12	13	14	0.3421	Not Spam	Contains standard conversational/business terms: [test, automated, api, report, suite]. Legitimate confidence score is …	2026-10-05 …
13	15	19	0.3421	Not Spam	Contains standard conversational/business terms: [test, automated, api, report, suite]. Legitimate confidence score is …	2026-10-05 …
14	18	24	0.3421	Not Spam	Contains standard conversational/business terms: [test, automated, api, report, suite]. Legitimate confidence score is …	2026-10-05 …
15	20	27	0.3553	Not Spam	Contains standard conversational/business terms: [test, api, verification, email]. Legitimate confidence score is 0.6447.	2026-10-05 …
16	22	30	0.3421	Not Spam	Contains standard conversational/business terms: [test, automated, api, report, suite]. Legitimate confidence score is …	2026-10-05 …
17	24	33	0.2388	Not Spam	Contains standard conversational/business terms: [report, submission, submit, 2026]. Legitimate confidence score is …	2026-10-06 …

APPLICATION FEATURES

LOGIN

DASHBOARD

EMAIL MANAGEMENT

SEARCH

SPAM DETECTION

REPORTS

User registration and login

JWT authentication

Password hashing using bcrypt

Inbox / Sent / Spam / Trash

Compose email

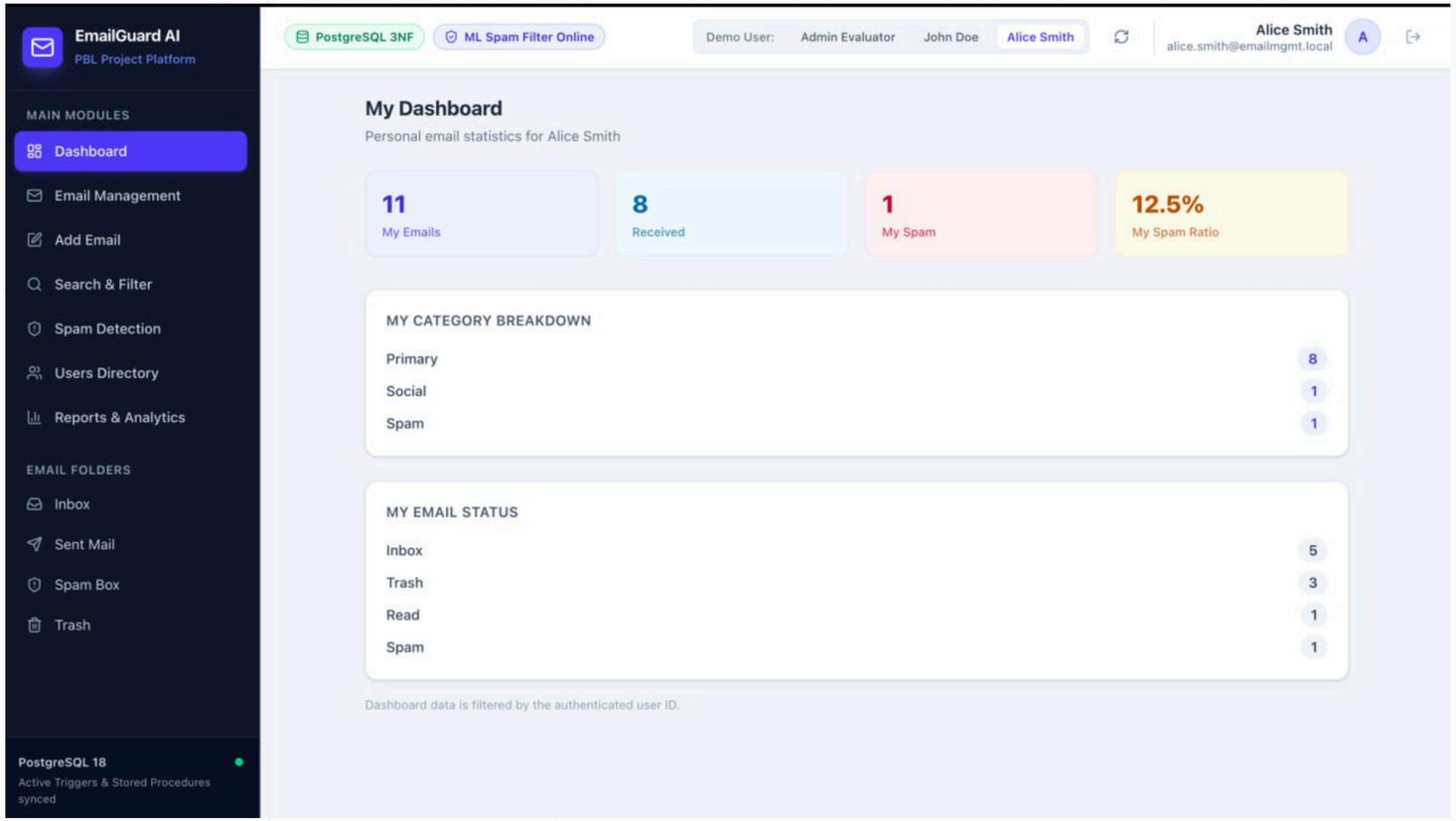
Email search

Spam prediction

Spam score and classification

Dashboard statistics

Reports and analytics



SYSTEM TESTING

USER

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FASTAPI

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POSTGRESQL

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SPAM CLASSIFIER

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SPAM RESULT

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FRONTEND

- ✓ Authentication
- ✓ User management
- ✓ Email management
- ✓ Email search
- ✓ Spam prediction
- ✓ Spam result storage
- ✓ Database triggers
- ✓ Stored functions
- ✓ Dashboard functionality
- ✓ Audit logging

	audit_id [PK] integ	email_id integer	action character varying (50)	previous_ characte	new_statu character	details text	changed_at timestamp with
1	1	1	CREATED	[null]	Inbox	Email created and delivered to receiver inbox	2026-10-05 ...
2	2	2	CREATED	[null]	Read	Email created and delivered to receiver inbox	2026-10-05 ...
3	3	3	CREATED	[null]	Inbox	Email created and delivered to receiver inbox	2026-10-05 ...
4	4	4	CREATED	[null]	Inbox	Email created and delivered to receiver inbox	2026-10-05 ...
5	5	5	CREATED	[null]	Inbox	Email created and delivered to receiver inbox	2026-10-05 ...
6	6	6	CREATED	[null]	Inbox	Email created and delivered to receiver inbox	2026-10-05 ...
7	7	7	CREATED	[null]	Inbox	Email created and delivered to receiver inbox	2026-10-05 ...
8	8	8	CREATED	[null]	Inbox	Email created and delivered to receiver inbox	2026-10-05 ...
9	9	9	CREATED	[null]	Inbox	Email created and delivered to receiver inbox	2026-10-05 ...
10	10	10	CREATED	[null]	Inbox	Email created and delivered to receiver inbox	2026-10-05 ...
11	11	11	CREATED	[null]	Inbox	Email created and delivered to receiver inbox	2026-10-05 ...
12	12	7	STATUS_CHANGED	Inbox	Spam	Status updated from Inbox to Spam	2026-10-05 ...
13	13	7	AUTO_SPAM_FLAGGED	Inbox	Spam	Automated ML Spam Classifier flagged message with confidence 98.0%	2026-10-05 ...
14	14	8	STATUS_CHANGED	Inbox	Spam	Status updated from Inbox to Spam	2026-10-05 ...
15	15	8	AUTO_SPAM_FLAGGED	Inbox	Spam	Automated ML Spam Classifier flagged message with confidence 96.0%	2026-10-05 ...
16	16	9	STATUS_CHANGED	Inbox	Spam	Status updated from Inbox to Spam	2026-10-05 ...
17	17	9	AUTO_SPAM_FLAGGED	Inbox	Spam	Automated ML Spam Classifier flagged message with confidence 92.0%	2026-10-05 ...
18	18	10	STATUS_CHANGED	Inbox	Spam	Status updated from Inbox to Spam	2026-10-05 ...
19	19	10	AUTO_SPAM_FLAGGED	Inbox	Spam	Automated ML Spam Classifier flagged message with confidence 97.0%	2026-10-05 ...
20	20	11	STATUS_CHANGED	Inbox	Spam	Status updated from Inbox to Spam	2026-10-05 ...
21	21	11	AUTO_SPAM_FLAGGED	Inbox	Spam	Automated ML Spam Classifier flagged message with confidence 94.0%	2026-10-05 ...
22	22	12	CREATED	[null]	Inbox	Email created and delivered to receiver inbox	2026-10-05 ...
23	23	12	STATUS_CHANGED	Inbox	Read	Status updated from Inbox to Read	2026-10-05 ...
24	24	13	CREATED	[null]	Inbox	Email created and delivered to receiver inbox	2026-10-05 ...
25	25	13	STATUS_CHANGED	Inbox	Spam	Status updated from Inbox to Spam	2026-10-05 ...

INSERT ACTUAL BACKEND TEST OUTPUT SCREENSHOT HERE

CONCLUSION & FUTURE SCOPE

CONCLUSION

Developed an integrated email management and spam detection platform.

PostgreSQL manages the relational data and DBMS logic.

FastAPI connects the application with the database and ML system.

Machine learning provides spam classification.

Spam results are stored and linked with email records.

React provides the user-facing application.

FUTURE SCOPE

Expand the dataset.

Improve spam classification.

Add larger-scale testing.

Improve analytics and visualization.

Deploy the system for production use.

THANK YOU